PUSHTI

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Exploring Substances : Acidic, Basic and Neutral

CHAPTER 01

Learning Outcomes...

- Acids : Their Properties and Uses
- What Do All Acids Have in Common?
- Bases : Their Properties and Uses
- What Do All Bases have in Common?
- Basicity of an Acid
- Acidity of a Base
- Indicators : Natural, Olfactory and Synthetic
- Neutralisation
- Importance of Neutralisation Reactions in Everyday Life
- Acid Rain
- Salts : Types, Properties and Uses

Introduction

Materials can be classified in a number of ways depending upon their characteristics. One such method classifies substances into acids, bases and salts. These have certain definite properties which distinguish one class from other. Acids and bases play a central role in chemistry. After studying reaction between acids and bases and their identification using indicator, we can use the processes in our daily life as well as for producing various useful products such as medicines and agricultural products, etc.

Acids

Don't forget.

- The substances which taste sour tend to contain acids in them and are called acidic in nature. The term 'acid' has been derived from the Latin word, *acere*, which means sour.
- Some of the acids are naturally occurring, which are generally present in food items. Lemon juice, orange juice, vinegar, curd, tamarind, etc. are sour in taste due to presence of acids.
- Following table shows the sources of few acids :

sour:

= organic acids

Sources	Acids
Lemon, orange	Citric acid
Vinegar	Acetic acid
Curd	Lactic acid
Tamarind	Tartaric acid
Apple	Maleic acid
Spinach, tomato	Oxalic acid
Ant sting, nettle sting	Methanoic acid

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Note: Acids from vegetation and animals is organic.

★ IMP -> LBH

Properties of Acids

ACIDS

- Acids have sour taste.
- Acids turn blue litmus red.
- Acids are soluble in water and produce a lot of heat when dissolved in water.
- Acidic solutions conduct electricity.
- They are corrosive. They destroy clothes, skin and paper.
- All acids generally contain hydrogen (hydrogen ions, H^+ ions).
- On reaction with metals most of the acids produce hydrogen gas.
- Acids react with bases to form salt and water. This reaction is called neutralisation reaction.
- Acids react with metal carbonates and metal hydrogen carbonates to form carbon dioxide, salt and water. Don't forget.

COMPETITION WINDOW

What Do All Acids Have in Common?

Generally all the acids contain hydrogen. The hydrogen present in acids is such that when acid dissolved in water it gives positively charged hydrogen ions (H^+ ions). Thus, acids can be defined as "A substance which dissociates on dissolving in water to produce hydrogen ions (H^+ ions)". pls remember!

$$\underset{\text{Hydrochloric acid}}{HCl_{(aq)}} \longrightarrow \underset{\text{Hydrogen ion}}{H^+_{(aq)}} + \underset{\text{Chloride ion}}{Cl^-_{(aq)}}$$

The separation of hydrogen ions from acids cannot occur in the absence of water.

- All hydrogen containing compounds are not acidic. Substances like ethyl alcohol (C_2H_5OH), glucose ($C_6H_{12}O_6$), etc. though contain hydrogen but when dissolved in water do not produce hydrogen ions (H^+ ions). Hence, they are not acidic.
- **Organic and Inorganic Acids :** Acids present in plants and animals (*i.e.*, living organisms) are called organic acids. These are weak acids, *e.g.*, citric acid, acetic acid, lactic acid, formic acid, etc.

  Acids that are obtained from minerals are called inorganic acids. These are strong acids, *e.g.*, hydrochloric acid (HCl), sulphuric acid (H_2SO_4), nitric acid (HNO_3), etc.

Uses of Acids

* IMP -> LBH USES

- Acids are used in laboratory as reagent.
- Sulphuric acid is called the 'King of Chemicals' because it is used in number of industries like fertilisers, drugs, plastics, paints, etc. It is also used in car batteries.
- Nitric acid is used for manufacturing explosives and fertilisers.
- Hydrochloric acid is used as a cleaner, to remove scales or deposits in the boilers.

DO YOU KNOW?

- There are few acids which play a very important role in our life.
  - Soaps and fats in our body contain acids which are called fatty acids.
  - Cells of our body contain an important acid DNA (deoxyribonucleic acid) which is responsible to pass characters from one generation to another.
  - Proteins are made up of acids called amino acids.
  - Hydrochloric acid is a strong acid which is found in our body in the gastric juice. It helps in digestion of food and kills harmful bacteria.
- The presence of acids has been detected on other planets as well. The atmosphere of planet Venus consists of thick white and pale yellow clouds of sulphuric acid.

Bases

BASE

- The substances which are bitter in taste and are soapy to touch contain bases in them and are called basic substances or bases.
- All bases are not soluble in water. The bases which are soluble in water are known as alkalies. Hence, all alkalies are bases but all bases are not alkalies. ★ Don't forget
- Following table shows examples of few bases with their chemical names :

Common Name	Chemical Name
Caustic soda	Sodium hydroxide, NaOH
Caustic potash	Potassium hydroxide, KOH
Milk of magnesia	Magnesium hydroxide, $Mg(OH)_2$
Slaked lime	Calcium hydroxide, $Ca(OH)_2$

DO YOU KNOW?

- Lime water (solution of calcium hydroxide in water) can be easily prepared by mixing quick lime (*chuna i.e.*, calcium oxide) in water and leaving it undisturbed for sometime. After filtration the liquid obtained is used as lime water. Just skim through
- Lime water is basic, while lime (which is a fruit similar to lemon) is acidic in nature.

Properties of Bases

★ IMP -> LBH

- Bases are bitter in taste.
- They turn red litmus blue.
- They possess a slippery or soapy touch.
- The solution of bases in water conduct electricity.
- Bases react with acids to form salt and water.

$$\underset{\substack{\text{Sodium hydroxide}\\ \text{(Base)}}}{2NaOH} + \underset{\substack{\text{Sulphuric acid}\\ \text{(Acid)}}}{H_2SO_4} \longrightarrow \underset{\substack{\text{Sodium sulphate}\\ \text{(Salt)}}}{Na_2SO_4} + \underset{\text{Water}}{2H_2O}$$

pls remember!

- Strong bases like caustic soda and caustic potash are corrosive in nature.

PROPERTIES

COMPETITION WINDOW

What Do All Bases have in Common?

When a base is dissolved in water, it always produces hydroxide ions (OH^- ions). Thus, a base is a substance which dissolves in water to produce hydroxide ions (OH^- ions) in solution. For example, sodium hydroxide is a base because it dissolves in water to produce hydroxide ions.

$$\underset{\substack{\text{Sodium hydroxide}\\ \text{(Base)}}}{NaOH_{(s)}} \xrightarrow{\text{Water}} \underset{\text{Sodium ion}}{Na^+_{(aq)}} + \underset{\text{Hydroxide ion}}{OH^-_{(aq)}}$$

Skim through.

Uses of Bases IMP → LBH

- Bases are used as common reagents in various industries.
- Calcium hydroxide is used for white washing of buildings.
- Sodium hydroxide is used for manufacturing soap. It is also used for manufacturing paper, rayon, textiles, etc.
- Ammonium hydroxide is used for manufacturing fertilisers, plastics, dyes, etc.
- Milk of magnesia (chemically magnesium hydroxide) is used as an antacid that acts as an agent to reduce acidity in stomach.

Ammonium hydroxide = NH_4OH

COMPETITION WINDOW

Types of Acids Remember

- **On the basis of strength :**
  - **Strong acids :** The acids, which when dissolved in water, get almost completely dissociated to provide a large amount of hydrogen ions are called strong acids, *e.g.*, sulphuric acid, hydrochloric acid, nitric acid, etc.
  - **Weak acids :** The acids, which when dissolved in water, are partially dissociated to provide a small amount of hydrogen ions are called weak acids, *e.g.*, acetic acid, phosphoric acid, etc. Organic
- **On the basis of amount of acid present :**
  - **Concentrated acids :** A concentrated acid is one which contains more amount of acid and less amount of water.
  - **Dilute acids :** A dilute acid is one which contains more amount of water and less amount of acid.

Basicity of an acid

- Number of replaceable hydrogen atoms present in a molecule of an acid is called basicity of an acid. Basicity of some common acids are :

Monobasic acid (Basicity - 1) : HCl (Hydrochloric acid)
Dibasic acid (Basicity - 2) : H_2SO_4 (Sulphuric acid)
Tribasic acid (Basicity - 3) : H_3PO_4 (Phosphoric acid)

Skim through.

Types of Bases Remember.

- **Strong bases :** A base which completely ionises in water and thus produces a large amount of hydroxide ions (OH^- ions) is called a strong base (or a strong alkali). For example, NaOH (commonly known as caustic soda), KOH (commonly known as caustic potash), etc.
- **Weak bases :** A base which partially ionises in water and thus produces a small amount of hydroxide ions is called a weak base. For example, $Mg(OH)_2$, $Ca(OH)_2$, NH_4OH, etc.

Acidity of a base

- Number of replaceable hydroxyl groups (OH^- ions) present in a base is called its acidity. Acidity of some common bases are :

Monoacidic base (Acidity - 1) : NaOH (Sodium hydroxide)
Diacidic base (Acidity - 2) : $Ca(OH)_2$ (Calcium hydroxide)
Triacidic base (Acidity - 3) : $Al(OH)_3$ (Aluminium hydroxide)

Skim through.

Indicators IMP → LBH

Remember and learn.

- An indicator is a substance which changes colour when added to an acid or a base. The change in colour indicates whether the substance is acidic or basic. If there is no change in the colour of the indicator, the substance is called a **neutral substance.**
- Indicators can be natural dyes extracted from plants or chemical compounds.

Skim

Natural Indicators

Litmus IMP - LBH.

- Litmus is a natural dye which is extracted from lichens. Litmus solution is prepared from this extract. It has a purple colour in distilled water. It turns red when added to an acidic solution and turns blue when added to a basic solution.
- It is also used in the form of litmus paper which is prepared by absorbing litmus solution on filter paper. Litmus is available in the form of blue litmus paper and red litmus paper.
- Acidic substances change blue litmus paper to red while basic substances change red litmus paper to blue.
- Substances which do not change the colour of the litmus paper are known as neutral substances.

DO YOU KNOW?

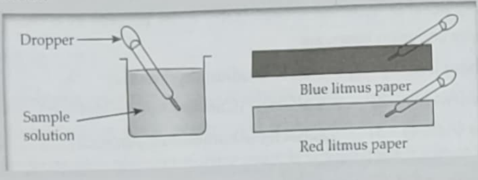
- Lichens are formed by the association of two living organism, a fungus and an algae. They grow on rocks and trees in region that have abundant rainfall and clean air.

ACTIVITY CORNER

To identify the given substances as acidic or basic in nature using litmus paper as indicator

Materials required : Vinegar, lemon juice, soap solution, soda water, curd, lime water, window cleaner, orange juice, aerated soft drink, sugar solution, salt solution, red litmus paper, blue litmus paper, dropper.

Procedure : Take a drop of the sample solution and put it on the blue litmus paper. Note the change in colour. Now put a drop of the same solution on red litmus paper and note the change in colour.



Observations :

Sample	Change in colour with blue litmus paper	Change in colour with red litmus paper
Vinegar	Red	No change
Lemon juice	Red	No change
Soap solution	No change	Blue
Soda water	Red	No change
Curd	Red	No change
Lime water	No change	Blue
Window cleaner	No change	Blue
Orange juice	Red	No change
Aerated soft drink	Red	No change
Sugar solution	No change	No change
Salt solution	No change	No change

Conclusion :

- Substances which change blue litmus paper to red are acidic substances.
- Substances which change red litmus paper to blue are basic substances.
- Substances which do not change the colour of litmus paper are neutral substances.

Red rose IMP – LBH

- Red rose indicator is prepared by soaking the petals of red rose in water.
- Red rose extract gives red colour in acidic solutions.
- Red rose extract gives green colour in basic solutions.
- Substances which do not changes the colour of red rose indicator are neutral in nature.

ACTIVITY CORNER

To identify the acidity or basicity of various substances using a red rose indicator

Materials required: Vinegar, lemon juice, soap solution, soda water, curd, lime water, window cleaner, orange juice, aerated soft drink, sugar solution, salt solution, red rose, warm water, dropper, mortar and pestle.

Procedure : Take some warm water in a beaker. Take some petals of red roses and wash them with water. Crush the petals using a mortar and pestle and place them in the beaker. Keep the petals dipped for some time till the water becomes coloured. Filter the coloured water and use it as an indicator by putting few drops in the sample solution. Observe the change in colour of the red rose extract with each sample.

Red rose

Red rose petals immersed in hot water

Observations :

Sample	Change in colour of red rose indicator
Vinegar	Red
Lemon juice	Red
Soap solution	Green
Soda water	Red
Curd	Red
Lime water	Green
Window cleaner	Green
Orange juice	Red
Aerated soft drink	Red
Sugar solution	No change
Salt solution	No change

Conclusion :

- Substances which turn red rose indicator to red are acidic in nature.
- Substances which turn red rose indicator to green are basic in nature.
- Sugar and salt solutions which do not show any colour change in the indicator are neutral.

China rose IMP – CBH

- China rose makes another natural indicator which is prepared by soaking the petals of the china rose in water.
- Acidic substances change China rose indicator into dark pink or magenta while basic substances turn China rose indicator into green.
- Substances which do not change the colour of China rose indicator are neutral in nature.

Turmeric IMP – LBH

- Turmeric or *haldi* powder that is commonly used in the kitchen is another natural indicator.
- Turmeric paper is made by depositing turmeric paste on blotting paper/filter paper and drying it.
- With acidic or neutral substances turmeric paper remains yellow.
- Turmeric paper changes to red colour in basic solutions.

DO YOU KNOW?

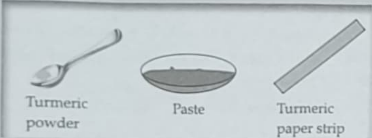
Turmeric (*curcuma longa*) is often used as an antiseptic for burns, cuts and stomach problems. Curcumin, the active ingredient in turmeric is being investigated for its potential to treat cancers, Alzheimer and arthritis.

ACTIVITY CORNER

To identify acidic or basic substances using turmeric powder as indicator

Materials required : Vinegar, lemon juice, soap solution, soda water, curd, lime water, window cleaner, orange juice, aerated soft drink, sugar solution, salt solution, turmeric powder, china dish, water, blotting paper

Procedure : Take one teaspoon of turmeric powder in china dish. Add a small amount of water to it to make a paste. Apply this paste on a blotting paper and let it dry. Cut the dried paper into thin strips. Dip the strip into each sample solution and note down the change in colour.



Observations :

Sample	Change in colour of turmeric paper
Vinegar	No change
Lemon juice	No change
Soap solution	Red
Soda water	No change
Curd	No change
Lime water	Red
Window cleaner	Red
Orange juice	No change
Aerated soft drink	No change
Sugar solution	No change
Salt solution	No change

Conclusion :

- With basic substances like soap solution, lime water and window cleaner, turmeric paper changes from yellow to red.
- With acidic or neutral substances turmeric paper does not change colour.

Red cabbage IMP - LBH.

- Red cabbage can also be used for making a natural indicator to test acids and bases.
- Red cabbage indicator can be prepared by boiling chopped red cabbage in water.
- Acidic solutions change purple colour of red cabbage indicator to red while basic solutions change purple colour of red cabbage to green.
- Neutral substances do not change the colour of red cabbage indicator.

DO YOU KNOW?

Hydrangea plant grows in cooler climates. It gives flowers of different colour, depending on the nature of soil. Acidic soil produces blue coloured flowers, whereas basic soil produces pink or red coloured flowers.

Olfactory Indicators

Those substances whose smell (or odour) changes in acidic or basic solutions are called olfactory indicators. An olfactory indicator usually works on the principle that when an acid or base is added to it then its characteristic smell cannot be detected. Onion, vanilla essence and clove oil are olfactory indicators.

These indicators give different odours in acidic and basic medium.

Indicator	Odour in acidic medium	Odour in basic medium
Onion extract	Characteristic ammoniacal smells	No smell
Vanilla essence	Characteristic sweet smell	No smell

IMP - LBH,

Synthetic Indicators

IMP - LBH.

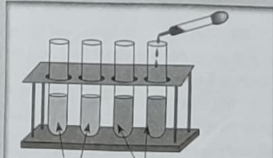
- Apart from the natural indicators, there are some synthetic indicators, which are prepared in laboratory, can also be used as indicators to identify acids and bases.
- Methyl orange and phenolphthalein are two common synthetic indicators used to test acids and bases.
- Methyl orange turns red in acidic solutions and yellow in basic solutions.
- Phenolphthalein remains colourless in acidic and neutral solutions and turns pink in basic solutions.

ACTIVITY CORNER

To identify the given samples of acid and base by using phenolphthalein and methyl orange as indicators

Materials required : Dilute hydrochloric acid, sodium hydroxide solution, phenolphthalein, methyl orange, dropper, test tubes

Procedure : Take a small amount of given solution in a test tube and add two drops of phenolphthalein. Repeat the experiment with methyl orange and record the observations.



Observations :

Sample	Indicator	Colour
Sodium hydroxide	Phenol-phthalein	Pink
Sodium hydroxide	Methyl orange	Yellow
Dilute hydrochloric acid	Phenol-phthalein	Colourless
Dilute hydrochloric acid	Methyl orange	Red

Conclusion :

- Phenolphthalein remains colourless in hydrochloric acid while turns pink with sodium hydroxide.
- Methyl orange turns yellow with sodium hydroxide while turns red with hydrochloric acid.

ILLUSTRATIONS

1 Why is an acid not stored in a metal container?

Ans.: Acids are corrosive in nature. They corrode metals because they react with metals to form salts and dissolve them. Hence, acids are stored in glass containers and not in metal containers.

2 To dilute an acid, always add acid to water and not water to acid. Why?

Ans.: When an acid is dissolved in water, heat is given out. If we add acid to water, the heat evolved is absorbed in water. If water is added to acid, large amount of heat is evolved and the acid starts boiling which can lead to accident. Hence, acids are diluted by adding acid to water slowly and with constant stirring.

3 What are acid-base indicators? Give examples of two natural and two synthetic indicators. Mention the colour change of these indicators with acids and bases.

Ans.: The substances which give different colours with acids and bases are called acid-base indicators. China rose and litmus are natural indicators. Methyl orange and phenolphthalein are synthetic indicators.

China rose	–	Magenta in acids and green in bases.
Litmus	–	Blue to red in acids and red to blue in bases.
Methyl orange	–	Orange to red in acids and orange to yellow in bases.
Phenolphthalein	–	Colourless in acids and pink in bases.

4 All alkalies are bases but all bases are not alkalies. Justify the statement.

Ans.: All bases are not soluble in water. The bases which are soluble in water and give hydroxide ions in the aqueous solutions are called alkalies. The bases which are not soluble in water are not alkalies. Hence, all alkalies are bases, but all bases are not alkalies.

5 What ions are present in solutions of following substances to predict they are acids or bases? (write symbols only).
Sulphuric acid, sodium hydroxide, copper hydroxide, acetic acid.

Ans.: Sulphuric acid	-	H^+ ions
Sodium hydroxide	-	OH^- ions
Copper hydroxide	-	OH^- ions
Acetic acid	-	H^+ ions

Neutralisation

HP+LBH vs

When acids and bases are mixed, they react and neutralise each other producing a salt which may be neutral, mildly acidic or basic. Water is also formed during the reaction and heat is evolved. The reaction between an acid and a base is called **neutralisation reaction**.

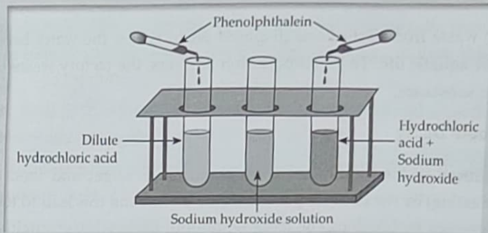
Acid + Base → Salt + Water + Heat

pls sum.

ACTIVITY CORNER

To study the reaction between an acid and a base (neutralisation reaction)

- **Materials required :** Dilute hydrochloric acid, dilute sodium hydroxide solution, phenolphthalein, test tubes, dropper
- **Procedure :** Take a test tube and add some dilute hydrochloric acid to it. Add few drops of phenolphthalein. Observe if there is any change in colour. Now add sodium hydroxide solution slowly. Keep shaking the test tube and observe change in colour. Keep adding sodium hydroxide till the solution becomes pink.
Add one more drop of sodium hydroxide and observe the colour of the solution. Touch the test tube and observe the temperature.



- **Observations :** When phenolphthalein is added to dilute hydrochloric acid, the solution is colourless. As we keep adding sodium hydroxide to it there is appearance of pink colour which disappears on shaking. When the acid and base neutralise each other, the resulting solution is neutral hence the solution is colourless. When an extra drop of sodium hydroxide is added, the solution turns pink. When we touch the test tube it feels warm.
- **Conclusion :** When a base and an acid are mixed in just right amount, neutralisation reaction takes place. When there is added extra base in the solution, it turns solution pink (as phenolphthalein turns pink in basic solutions).

Importance of Neutralisation Reactions in Everyday Life

Treatment of Acidity and Indigestion

IN your own words

- In our stomach, hydrochloric acid helps in digestion. But excess of acid in the stomach causes acidity and indigestion. It can be treated by neutralising the acid with a mild base. These bases are called antacids. Milk of magnesia is an antacid which contains a mild base, magnesium hydroxide.

Treatment of Acidity of Soil

- Due to use of excess of fertilisers in the soil, the nature of the soil becomes acidic. Acidic soil is not good for plants. To neutralise the acidity of the soil some bases like slaked lime (calcium hydroxide) or quick lime (calcium oxide) is added to the soil.
- If the soil is basic, organic matter (compost) is added to it. Organic matter releases acids which neutralises the basic nature of the soil.

Treatment of Insect Bite

- The sting of an ant contains formic acid which causes pain and inflammation. To get relief from ant sting the acid is neutralised by bases like sodium bicarbonate (baking soda) or zinc carbonate (calamine solution).
- Wasp sting on the other hand, is basic and vinegar (which contains acetic acid) is used for relief from wasp sting.

Skim through

Treatment of Factory Wastes

IN your own words

- When acidic waste from factories is disposed off in rivers, the water becomes acidic and is harmful for aquatic life. To avoid pollution of rivers, the factory waste is neutralised by adding basic substance.

Prevention of Tooth Decay

- Acid is produced in the mouth due to degradation of sugar and food particles (left in mouth after eating) by the bacteria present in the mouth and this lead to tooth decay. Tooth decay is prevented by brushing the teeth with tooth paste which contains bases, resulting in neutralisation of the acid thus preventing cavities.

Skim through

COMPETITION WINDOW

Acid Rain

- The rain which contains excess of acids dissolved in it, is called acid rain.
- The acidic oxides present in the air like carbon dioxide, sulphur dioxide and nitrogen dioxide are dissolved in rain water to form acids like carbonic acid, sulphuric acid and nitric acid.
- Acid rain causes damage to crops, soil and buildings made up of marbles.

pH

- The strength of an acid and a base is measured by a scale called pH scale. pH value of solution indicates the concentration of hydrogen ions in the solution. Pure water which is neutral has a pH value 7. A pH paper is used to find out pH of a solution by matching the change in colour with a universal indicator paper.
- In general, lesser the pH of a solution, more will be its acidic strength. Similarly, higher the pH of a solution, more will be its basic strength.
  - A neutral solution has pH = 7.
  - An acidic solution has pH < 7.
  - A basic solution has pH > 7.
- A pH plays vital role in our daily life because most of the biochemical reactions take place at specific pH values. pH of our blood is in the range of 7.35 – 7.45. Our body becomes prone to diseases when pH of blood alters due to some reason.

Salts

* IMP – LBH

Salts are formed when acids react with bases. So, the name of a salt consists of two parts: the first part of the name of salt is derived from the name of the base, and the second part of the name of the salt comes from name of acid. For example, sodium chloride is a salt, comes from sodium hydroxide and hydrochloric acid.

Types of Salts

* IMP – LBH.

- **Neutral salts :** Salts formed by a strong acid and a strong base are called neutral salts. Neutral salts produce neutral solutions when dissolved in water, *e.g.*, sodium chloride (salt of HCl and NaOH).

strong Acid + weak base

- **Acidic salts :** Salts formed by neutralisation of a strong acid and a weak base give acidic salts, *e.g.*, ammonium chloride (salt of HCl and NH_4OH).
- **Basic salts :** Salts formed by neutralisation of a strong base and a weak acid give basic salts, *e.g.*, sodium acetate (salt of NaOH and CH_3COOH).

strong bases + weak acid.

ACTIVITY CORNER

To demonstrate that a salt is formed by the reaction between an acid and a base

- **Materials required :** Hydrochloric acid (HCl), copper oxide (CuO), beaker, glass rod
- **Procedure :** Take 2 g of black CuO (base) in a beaker and add few drops of HCl (acid) in it and stir the contents with the help of glass rod. Observe the colour changes.
- **Observations :** After adding HCl the black colour of CuO changes to bluish-green colour.
- **Conclusion :** When HCl (acid) reacts with CuO (base) a bluish-green colour of $CuCl_2$ (salt) is formed.

$$\underset{\text{(Black)}}{CuO_{(s)}} + 2HCl_{(aq)} \rightarrow \underset{\text{(Bluish green)}}{CuCl_{2(aq)}} + H_2O$$

COMPETITION WINDOW

Properties of Salts

- **Reaction with an acid :** When a salt reacts with an acid, another salt and acid are formed.

$$NaCl + H_2SO_4 \longrightarrow NaHSO_4 + HCl$$
$$2NaCl + H_2SO_4 \longrightarrow Na_2SO_4 + 2HCl$$

Skim through

- **Reaction with a base :** A salt reacts with a base to produce another salt and base.

$$(NH_4)_2SO_4 + 2NaOH \longrightarrow Na_2SO_4 + 2NH_4OH$$

- **Reaction with a metal :** Sometimes, a salt solution may react with a metal. For example, when an iron nail is dipped into an aqueous solution of copper sulphate, copper gets deposited on the surface of the nail and the ferrous sulphate formed remains in the solution.

$$CuSO_4 + Fe \longrightarrow FeSO_4 + Cu$$

Uses of Salts

- Sodium chloride (common salt) is an important part of our food and vital to health. It is used as a preservative, in making soaps and preparing many chemicals.
- Sodium carbonate (washing soda) is an important part of detergents. It is also used in making glass.
- Sodium bicarbonate (baking soda) is used in baking cakes and bread. It is also used in fire extinguishers.
- Copper sulphate (blue vitriol) is used as a fungicide and in electroplating.

$CuSO_4$

DO YOU KNOW?

- Soap is manufactured by reaction of oils with alkalies. The soap formed remains soluble in the solution. On stirring with common salt, soap solidifies out. This is called **salting out of soap**.
- The fixed number of water molecules which remain with the crystalline salts is known as **water of crystallisation**, *e.g.*, $FeSO_4 \cdot 7H_2O$, $CuSO_4 \cdot 5H_2O$, $Na_2CO_3 \cdot 10H_2O$. Such water molecules give shape and colour to the crystals. Salts with water of crystallisation are called **hydrated salts**. On heating, hydrated salts lose their shape and colour and change into a powdery substance. These salts are called **anhydrous salts**.
- Some salts on exposure to atmosphere absorb moisture and become wet. Such salts are called deliquescent salts. Some crystalline hydrated salts on exposure to atmosphere lose their water of crystallisation. Such salts are called **efflorescent salts.**

ILLUSTRATIONS

6 What happens when dilute sulphuric acid and lime water are mixed together?

Ans.: When sulphuric acid (H_2SO_4) and lime water ($Ca(OH)_2$) are mixed, neutralisation reaction takes place. The salt formed is calcium sulphate ($CaSO_4$). Water (H_2O) and heat are produced as a result of neutralisation.

$$\underset{Ca(OH)_2}{\text{Lime water}} + \underset{H_2SO_4}{\text{Sulphuric acid}} \rightarrow \underset{CaSO_4}{\text{Calcium sulphate}} + \underset{2H_2O}{\text{Water}}$$

7 Sodium hydroxide is taken in a test tube and few drops of phenolphthalein are added to it. Then dilute hydrochloric acid is added dropwise. What will you observe and why? What will happen if an excess of acid is added?

Ans.: When phenolphthalein is added to sodium hydroxide solution, pink colour appears. When hydrochloric acid is added to it, the pink colour disappears because neutralisation reaction takes place. Sodium hydroxide is neutralised by hydrochloric acid. If we keep on adding hydrochloric acid even after decolourisation of solution, it will remain colourless because acids do not change the colour of phenolphthalein.

8 Give name and formula of any two salts.

Ans.: Sodium chloride, NaCl

Copper sulphate, $CuSO_4$

9 Potassium nitrate is a salt. Give names of acid and base used to form this salt.

Ans.: Potassium nitrate (KNO_3) is formed from potassium hydroxide (base) and nitric acid (acid).

$$\underset{\substack{\text{Potassium}\\\text{hydroxide}\\\text{(Base)}}}{KOH} + \underset{\substack{\text{Nitric}\\\text{acid}\\\text{(Acid)}}}{HNO_3} \longrightarrow \underset{\substack{\text{Potassium}\\\text{nitrate}\\\text{(Salt)}}}{KNO_3} + H_2O$$

10 Explain why aqueous solution of sodium chloride is neutral.

Ans.: As sodium chloride is formed by the combination of a strong acid (HCl) and strong base (NaOH), thus it is a neutral salt.

slake lime = Ca(OH)
↳ calcium hydroxide

CONCEPT MAP

Exploring Substances : Acidic, Basic and Neutral

Acids

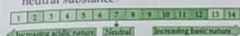
- An acid is a chemical substance that has a sour taste.
- Turns blue litmus red.
- Generally corrosive in nature.
- Gives H^+ ions in aqueous solutions.
- May be mineral (inorganic) or organic.

Uses

- As a preservative in food.
- In the manufacture of fertilisers, plastics, etc.
- In lead storage batteries.
- As a laboratory reagent.

Strength of Acids or Bases

- The acids which get completely ionised in water to give H^+ ions are called strong acids.
- The bases which get completely ionised in aqueous solution to give OH^- ions are called strong bases.
- Strength of an acidic or basic solution can be measured by using pH scale.
- The pH value ranges from 1 to 14 with 1 being the most acidic substance and 14 being the most basic substance while 7 is the neutral substance.



Bases

- A base is a chemical substance that has bitter taste.
- Turns red litmus blue.
- Are soapy to touch.
- Water soluble bases are called alkalies.
- Gives OH^- ions in aqueous solutions.

Uses

- In manufacture of soap, paper, etc.
- As antacids (*e.g.*, milk of magnesia).
- In window cleaners.
- As a laboratory reagent.

Indicators

- An indicator is a substance that can determine whether the substance is acidic or basic.
- Litmus, turmeric, Red rose, China rose and red cabbage are examples of natural indicators.
- Phenolphthalein and methyl orange are examples of synthetic indicators.
- Some common indicators and their colour change :

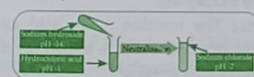
Indicators	Colour in acid	Colour in base
Litmus paper	Blue litmus turns red	Red litmus turns blue
Red cabbage (purple)	Red	Green
Turmeric (yellow)	Yellow	Red
China rose (light pink)	Dark pink (Magenta)	Green
Red rose	Red	Green
Phenolphthalein (colourless)	Colourless	Pink
Methyl orange (orange)	Red	Yellow

Salts

- When a base is added to an acid, neutralisation occurs and a salt is formed.
- **Normal salts** : Are neutral salts, *e.g.*, NaCl, KCl, etc.
- **Acidic salts** : Are salts formed by reaction between a strong acid and a weak base, *e.g.*, NH_4Cl, $CuSO_4$, etc.
- **Basic salts** : Are salts formed by reaction between a strong base and a weak acid, *e.g.*, Na_2CO_3, CH_3COONa, etc.

Neutralisation Reactions

- Neutralisation is a process or a chemical reaction in which an acidic and basic substance is mixed with each other in order to neutralise their acidic or alkaline nature.
- Always evolve heat.



Neutralisation Reactions in Everyday Life

- Farmers use lime to neutralise acidic soil and organic matter to neutralise basic soil.

- Our stomach contains hydrochloric acid and too much of this causes indigestion.

Antacid tablets contain bases such as magnesium hydroxide and magnesium carbonate to neutralise the extra acid.

- Bee stings are acidic(contain formic acid). They can be neutralised using baking powder (sodium hydrogen carbonate) and calamine lotion (zinc carbonate).

- Acids are responsible for tooth decay. Toothpastes helps in neutralising the effect of acid because they are alkaline in nature and neutralise the effect of acid.

Solved Examples

1. Baking soda is used to treat bee sting while vinegar is used to treat wasp sting. Based on this information identify the difference in chemical nature of wasp sting and ant sting.

Ans.: Since, baking soda is used to treat bee sting it means bee sting contains acid which is neutralised by baking soda. Since vinegar is used to treat wasp sting it indicates that wasp sting contains some base which is neutralised by vinegar.

2. Name three acids obtained from natural sources.

Ans.: (i) Citric acid from oranges
(ii) Tartaric acid from tamarind
(iii) Oxalic acid from tomatoes

3. Why does a milkman usually add a very small amount of baking soda to fresh milk during summer season?

Ans.: Baking soda is basic in nature. It neutralises the lactic acid present in milk and prevents it from turning sour.

4. What is a neutralisation reaction? Give two examples.

Ans.: The reaction of an acid and a base to form salt and water is known as neutralisation reaction. Examples :

(i) $\underset{\text{Sodium hydroxide}}{NaOH} + \underset{\text{Hydrochloric acid}}{HCl} \rightarrow \underset{\text{Sodium chloride}}{NaCl} + \underset{\text{Water}}{H_2O}$

(ii) $\underset{\text{Potassium hydroxide}}{2KOH} + \underset{\text{Sulphuric acid}}{H_2SO_4} \rightarrow \underset{\text{Potassium sulphate}}{K_2SO_4} + \underset{\text{Water}}{2H_2O}$

5. Define acidity of a base. Give one example each of monoacidic, diacidic and triacidic bases.

Ans.: Acidity of a base : Number of replaceable hydroxyl groups (OH^- ions) present in a base is called its acidity.

Monoacidic base : $NaOH$ (Sodium hydroxide)
Diacidic base : $Ca(OH)_2$ (Calcium hydroxide)
Triacidic base : $Al(OH)_3$ (Aluminium hydroxide)

6. What will you observe when
(i) methyl orange is added to dilute hydrochloric acid?
(ii) a drop of phenolphthalein is added to lime water?

Ans.: (i) Methyl orange will change to red in acidic solution.
(ii) The colour of phenolphthalein will become pink in basic solution as lime water is basic in nature.

7. Why should a farmer add quick lime to the soil if he thinks his crop is not growing well?

Ans.: The farmer should add quick lime to the soil if the soil condition is more acidic than the required conditions. Quick lime will neutralise the excess of acid present in soil.

8. Why is sodium hydrogencarbonate called baking soda and sodium carbonate as washing soda?

Ans.: Sodium hydrogencarbonate is a mild base and is used for making baking powder which is used in bakery for making cookies, cakes, etc. That is why it is called baking soda.
Sodium carbonate is used in laundries for cleaning of clothes, hence it is called washing soda.

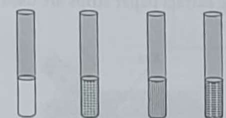
9. Name the base used in manufacture of soap.

Ans.: Sodium hydroxide is used to manufacture soap. Soaps are sodium salts of fatty acids.

10. You are provided with three test tubes containing three colourless liquids namely water, vinegar and caustic soda. How will you identify the liquids?

Ans.: (i) Add blue litmus paper in all the three test tubes. The test tube in which the litmus paper turns red contains vinegar.
(ii) Add red litmus paper in the remaining two test tubes. The solution which turns red litmus blue contains caustic soda.
(iii) The solution which does not change the colour of the red and blue litmus paper is water.

11. Look at figure which shows solutions taken in test tubes *A*, *B*, *C* and *D*. What colour is expected when a piece of red litmus paper is dropped in each test tube? Nature of the solutions is given in the table for your help.



Test tube	Nature of solution	Change in colour of red litmus
A	Neutral	
B	Basic	
C	Acidic	
D	Neutral	

(NCERT Exemplar)

Ans.:

Test tube-solution in water	Nature of solution	Change in colour of red litmus
A (Milk)	Neutral	No change
B (Baking powder)	Basic	Turns blue
C (Vinegar)	Acidic	No change
D (Glucose)	Neutral	No change

12. Paheli observed that most of the fish in the pond of her village were gradually dying. She also observed that the waste of a factory in their village is flowing into the pond which probably caused the fish to die.
(a) Explain why the fish were dying.
(b) If the factory waste is acidic in nature, how can it be neutralised? *(NCERT Exemplar)*

Ans.: (a) Since the factory waste is being disposed off in the river, it can kill the fish because factory waste may contain acids or bases or other toxic materials.
(b) If the factory waste is acidic in nature, it can be neutralised by adding basic substances like lime before disposing off in the river.

13. While playing in a park, a child was stung by a wasp. Some elders suggested applying paste of baking soda and others lemon juice as remedy. Which remedy do you think is appropriate and why? *(NCERT Exemplar)*

Ans.: Wasp sting injects a liquid in the skin which is acidic in nature. Hence, baking soda is applied as it is basic in nature and neutralises the acid.

14. Boojho, Paheli and their friend Golu were provided with a test tube each containing China rose solution which was pink in colour. Boojho added two drops of solution '*A*' in his test tube and got dark pink colour. Paheli added 2 drops of solution '*B*' to her test tube and got green colour. Golu added 2 drops of solution '*C*' but could not get any change in colour. Suggest the possible cause for the variation in their results. *(NCERT Exemplar)*

Ans.: '*A*' is acidic solution, China rose solution turns dark pink in acids. '*B*' is a basic solution since China rose solution turns green in basic solution. '*C*' is a neutral solution hence there is no change in colour of China rose solution.

NCERT Section

Let us Enhance Our Learning

1. A solution turns the red litmus paper to blue. Excess addition of which of the following solution would reverse the change?
   (i) Lime water (ii) Baking soda
   (iii) Vinegar
   (iv) Common salt solution

2. You are provided with three unknown solutions labelled *A*, *B*, and *C*, but you do not know which of these are acidic, basic, or neutral. Upon adding a few drops of red litmus solution to solution *A*, it turns blue. When a few drops of turmeric solution are added to solution *B*, it turns red. Finally, after adding a few drops of red rose extract to solution *C*, it turns green.
   Based on the observations, which of the following is the correct sequence for the nature of solutions *A*, *B*, and *C*?
   (i) Acidic, acidic, and acidic
   (ii) Neutral, basic, and basic
   (iii) Basic, basic, and acidic
   (iv) Basic, basic, and basic

3. Observe and analyse Figs. 1, 2, and 3, in which red rose extract paper strips are used. Label the nature of solutions present in each of the containers.

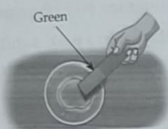


Fig. 1

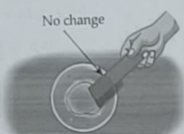


Fig. 2

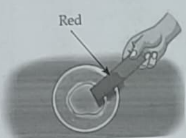


Fig. 3

4. A liquid sample from the laboratory was tested using various indicators:

Indicator	Red litmus	Blue litmus	Turmeric
Change	No change	Turned red	No change in colour

Based on the tests, identify the acidic or basic nature of the liquid and justify your answer.

5. Manya is blindfolded. She is given two unknown solutions to test and determine whether they are acidic or basic. Which indicator should Manya use to test the solutions and why?

6. Could you suggest various materials which can be used for writing the message on the white sheet of paper (given at the beginning of the chapter) and what could be in the spray bottle? Make a table of various possible combinations and the colour of the writing obtained.

7. Grape juice was mixed with red rose extract; the mixture got a tint of red colour. What will happen if baking soda is added to this mixture? Justify your answer.

8. Keerthi wrote a secret message to her grandmother on her birthday using orange juice. Can you assist her grandmother in revealing the message? Which indicator would you use to make it visible?

9. How can natural indicators be prepared? Explain by giving an example.

10. Three liquids are given to you. One is vinegar, another is a baking soda solution, and the third is a sugar solution. Can you identify them only using turmeric paper? Explain.

11. The extract of red rose turns the liquid *X* to green. What will the nature of liquid *X* be? What will happen when excess of amla juice is added to liquid *X*?

12. Observe and analyse the information given in the following flowchart. Complete the missing information.

Imagine a garden with plants showing signs of poor health.

The soil can be ______ in nature. | The soil can be ______ in nature.

Which indicator can be used to test the nature of the soil?
______

The acidic soil can be treated with ______. | The basic soil can be treated with ______.

Exercise

Multiple Choice Questions

LEVEL - 1

Acids, Bases and Indicators

1. Among lemon juice and coffee,
   (a) both are acidic
   (b) both are basic
   (c) lemon juice is acidic, coffee is basic
   (d) lemon juice is basic, coffee is acidic. *(NCERT Page 10)* U

2. Which of the following will turn blue litmus red?
   (a) Lime water (b) Lemon water
   (c) Sugar solution (d) Salty water *(Page 10)* U

3. Vinegar is
   (a) bitter in taste (b) sweet in taste
   (c) sour in taste (d) tasteless. *(Page 11)*

4. Which of the following acids is present in curd?
   (a) Lactic acid (b) Formic acid
   (c) Acetic acid (d) Citric acid *(Page 11)* R

5. Acids present in fruits and vegetables are called
   (a) organic acids (b) mineral acids
   (c) strong acids (d) fruit acids. R

6. The main source of oxalic acid is
   (a) curd (b) spinach
   (c) lemon (d) orange.

7. To dilute a concentrated acid
(a) both acid and water are mixed in same amount.
(b) water is added to the acid in one go.
(c) acid is added to the water slowly.
(d) base is added to the acid slowly. An

8. Which of the following is a mineral acid?
(a) Lactic acid (b) Formic acid
(c) Tartaric acid (d) Nitric acid U

9. Which of the following explains the properties—corrosive, sour, water soluble, present in citrus fruits?
(a) Acid (b) Base
(c) Salt (d) Alkali
(Page 11) U

10. Which of the following is not an inorganic acid?
(a) Hydrochloric acid (b) Nitric acid
(c) Sulphuric acid (d) Acetic acid R

11. Citric acid is present in
(a) curd (b) milk
(c) lemon (d) spinach.
(Page 11) R

12. When phenolphthalein is added to acids, its colour
(a) changes to pink
(b) changes to green
(c) remains colourless
(d) changes to orange.

13. Lemon juice will turn
(a) phenolphthalein pink
(b) red litmus blue
(c) turmeric indicator red
(d) methyl orange red. U

14. Which of the following is not the property of an alkali?
(a) They have bitter taste.
(b) They turn red litmus blue.
(c) They are insoluble in water.
(d) They give pink colour with phenolphthalein. R

15. A substance that changes colour in acidic and basic solutions is called
(a) an indicator (b) a weak base
(c) a weak acid (d) a neutral salt.
(Page 11) R

16. The common window cleaner contains
(a) calcium hydroxide
(b) magnesium hydroxide
(c) sodium hydroxide
(d) ammonium hydroxide. R

17. Add few drops of red rose indicator in soap solution. What do you observe?
(a) It turns green. (b) It turns magenta.
(c) It turns yellow. (d) It turns red.
(Page 12)

18. Add a few drops of methyl orange in caustic soda solution. What do you observe?
(a) It turns yellow.
(b) It turns red.
(c) It remains orange.
(d) It becomes colourless.

Neutralisation Reaction and its Importance in Everyday Life, Salts

19. When an acid and a base react, salt (X) and another product (Y) are formed. (Y) is
(a) an acid (b) a base
(c) water (d) hydrogen.
(Page 18) An

20. A test tube contains pink solution formed by adding few drops of phenolphthalein to sodium hydroxide solution. Few drops of sulphuric acid are added to the test tube. It is observed that the pink colour of the solution
(a) turns dark pink
(b) becomes colourless
(c) turns purple
(d) does not change. U

21. Acidic soil can be neutralised by
(a) vinegar (b) quick lime
(c) orange juice (d) tamarind juice.
(Page 18) AP

22. Which of the following statements is not correct?
(a) Acids turn blue litmus red.
(b) Bases are bitter in taste.
(c) Neutralisation reaction gives salt and water.
(d) All the indicators change colour in acids and bases. An

23. Hydrochloric acid can be neutralised by
(a) sodium hydroxide
(b) calcium hydroxide
(c) sodium carbonate
(d) all of these.

24. Which of the following types of medicines is used for treating indigestion?
(a) Antibiotic (b) Antacid
(c) Antiseptic (d) Antipyretic
Ap

25. Which of the following is not correctly matched?
(a) Substances that turn red litmus blue – Bases
(b) Substances used for testing acids and bases – Indicators
(c) An example of neutral substance – Common salt
(d) Substance used for treatment of ant bite – Acetic acid
An

26. If the soil is too basic, it can be neutralised by adding
(a) lime
(b) baking powder
(c) manure
(d) chalk powder. (Page 19) Ap

27. Calamine solution is applied on the area of ant bite. Calamine solution contains
(a) sodium carbonate
(b) sodium hydrogencarbonate
(c) zinc carbonate
(d) sodium chloride. R

28. Common name of sodium bicarbonate is
(a) common salt (b) baking soda
(c) washing soda (d) quick lime. R

29. The acidic waste of factories, before disposing off in the river is treated with
(a) hydrochloric acid (b) bleaching powder
(c) bases (d) salts. (Page 19)

30. The acid present in the stomach which helps in digestion is
(a) sulphuric acid
(b) citric acid
(c) lactic acid
(d) hydrochloric acid. R

LEVEL - 2

31. A boy was given two test tubes, one containing water and other containing sodium hydroxide. To identify the solutions which of the following indicators can be used by him?
(i) Blue litmus
(ii) Turmeric indicator
(iii) Phenolphthalein
(iv) Red litmus
(a) (i) and (iv) (b) (ii), (iii) and (iv)
(c) only (iv) (d) only (iii) Ap

32. Neutralisation is used in the treatment of
(a) tooth decay
(b) indigestion
(c) excess acidity or basicity of soil
(d) all of these. Ap

33. Which of the following statements is correct?
(a) China rose indicator turns window cleaner solution orange.
(b) Soda water will turn green when red cabbage juice is added to it.
(c) It we put a drop of soap solution on the strip of the turmeric paper it will turn red.
(d) None of these

34. Which of the following is a strong acid?
(a) Hydrochloric acid (b) Carbonic acid
(c) Acetic acid (d) Lactic acid R

35. Which of the following will not change the colour of turmeric solution?
(a) Baking soda (b) Lime water
(c) Soap solution (d) Salt solution
Ap

36. Which of the following acids are not present in acid rain?
(a) Sulphuric acid
(b) Nitric acid
(c) Carbonic acid
(d) Hydrochloric acid R

37. Aqueous solution of sodium chloride
(a) turns red litmus blue
(b) turns blue litmus red
(c) turns red litmus orange
(d) does not change the colour of either red or blue litmus. U

38. A milkman added a small pinch of baking soda to fresh milk which had pH close to 6. As a result, pH of the medium
(a) became close to 2
(b) became close to 4
(c) did not undergo any change
(d) became close to 8. U

39. Which of the following groups of samples will not show any change in colour with China rose indicator?
(a) Lemon juice, curd and sugar solution
(b) Distilled water, sodium chloride solution and sugar solution
(c) Distilled water, window cleaner and lime water
(d) Lime water, soap solution and lemon juice

40. Molecular formulae of baking soda and washing soda are respectively
(a) Na_2CO_3, $NaHCO_3$
(b) $NaCl$, Na_2CO_3
(c) $NaHCO_3$, Na_2CO_3
(d) $NaOH$, $NaHCO_3$ R

41. A set of students went on a nature trip where one of the students disturbed the honey comb by throwing a stone on it. Few students were stung by the bee. A person gathering medicinal plants, came to their rescue and applied the extract of some leaves, which relieved the students of their pain. The chemical nature of leaf would have been
(a) acidic (b) basic
(c) neutral (d) mildly acidic. Ap

42. Two solutions A and B have pH value 2 and 5 respectively, their nature will be as follows
(a) A and B both acidic
(b) A alkaline, B acidic
(c) B alkaline, A acidic
(d) A and B both alkaline. An

43. Identify the wrong statement.
(a) Bases are corrosive and have sour taste.
(b) Milk of magnesia contains magnesium hydroxide.
(c) Sodium hydroxide is a strong base.
(d) Water is produced during neutralisation. An

44. Which of the following is not an important use of the acids?
(a) To manufacture fertilisers
(b) In textile, leather and paper industry
(c) In white washing of buildings
(d) In removing scales from boilers Ap

45. Sodium hydroxide is not used for neutralising the hydrochloric acid present in the stomach during acidity because
(a) it is a weak base and cannot neutralise acid
(b) it is a strong base and it will burn the mouth and walls of stomach
(c) it is not soluble in water hence cannot be swallowed
(d) the acid present in stomach does not react with sodium hydroxide. U

46. Riya prepared an indicator paper by dipping a paper strip in a solution of X. She put a few drops of an unknown solution Y (in which number of H^+ ions is less than that of OH^- ions) on this paper. She observed that the colour of paper turned yellow. Solutions X and Y could be respectively
(a) Red litmus solution and tamarind juice
(b) Phenolphthalein and baking soda solution
(c) Methyl orange and lime water
(d) Turmeric solution and lemon juice Ap

47. Acids and bases should not be stored in metal containers because
(a) they are corrosive in nature and react with metals
(b) they cannot be seen from outside
(c) they become brown in colour when stored in metal containers
(d) metal containers are heavy and cannot be lifted when filled with acids and bases. U

48. Both lime water and sodium hydroxide are ______ in nature. They do not turn ______.
(a) acidic, red litmus blue
(b) basic, phenolphthalein pink
(c) acidic, methyl orange yellow
(d) basic, turmeric solution green. U

49. Which of the following is/are acidic in nature?
(I) Lime juice (II) Human blood
(III) Lime water (IV) Vitamin C tablet
(a) Only I and IV (b) Only III and IV
(c) Only IV (d) I, II, III and IV U

50. If a few drops of a concentrated acid accidentally spill over the hand of a student, what should be done?
(a) Wash the hand with saline water.
(b) Wash the hand immediately with plenty of water and apply a paste of sodium hydrogencarbonate.
(c) After washing with plenty of water apply solution of sodium hydroxide on the hand.
(d) Neutralise the acid with a strong alkali. Ap

LEVEL - 3 (HOTS)

Competency Focused Questions (CFQs)

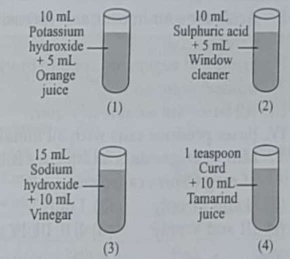
51. Which of the following groups of acids contains only mineral acids?
(a) Citric acid, Formic acid, Nitric acid
(b) Acetic acid, Ascorbic acid, Formic acid
(c) Hydrochloric acid, Sulphuric acid, Nitric acid
(d) Citric acid, Lactic acid, Sulphuric acid

52. Archi, a class 7 student tabulated the nature of a few substances with common indicators in the given table.

S. No.	Substance	Colour with China rose indicator	Colour with methyl orange indicator
I.	Nitric acid	Green	Yellow
II.	Soda water	Magenta	Red
III.	Milk of magnesia	Magenta	Red
IV.	Spinach juice	Magenta	Red
V.	Lime water	Magenta	Red

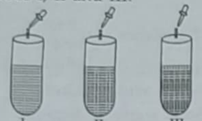
The incorrect entries are
(a) II and V only (b) I and III only
(c) I, III and V only (d) II and III only. An

53. A class 7 science teacher arranged the following set of test tubes.



Which of the following observations is correct when these solutions were tested with different indicators?
(a) In test tubes 1 and 3, turmeric solution remains yellow.
(b) In test tubes 2 and 4, China rose indicator turns magenta.
(c) Methyl orange turns red in test tubes 1 and 3.
(d) Phenolphthalein turns pink in test tubes 2 and 4. Cr

54. Given figure shows the colour changes when China rose indicator is added in test tubes I, II and III. Identify the solutions present in test tubes I, II and III.



	I	II	III
(a)	Sugar solution	Lime water	Baking powder
(b)	Sugar solution	Lemon juice	Vinegar
(c)	Lime water	Sugar solution	Lemon juice
(d)	Lemon juice	Sugar solution	Lime water

55. Which of the following statements is/are correct?

I. Acids are sour in taste and corrosive in nature.
II. Fruits and vegetables contain inorganic acids.
III. All bases are soluble in water.
IV. Bases produce salts with all metals.
V. Milk of magnesia is an aqueous solution of magnesium carbonate.

(a) II and IV only (b) I only
(c) III and V only (d) I, II, III, IV and V

56. A few solutions were tested with different indicators and results were noted down in the given table.

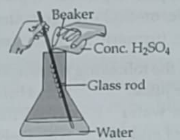
S. No.	Solution	Colour with China rose indicator	Colour with methyl orange indicator
1.	V	Dark pink	Red
2.	W	Dark pink	Red
3.	X	No change	No change
4.	Y	Green	Yellow
5.	Z	Light pink	Orange

Study the given table carefully and fill in the blanks by choosing an appropriate option. Solutions (i) will have no effect on blue litmus paper while solution(s) (ii) will turn blue litmus red. Phenolphthalein remains colourless in solution(s) (iii) while it changes to pink colour in solution(s) (iv).

	(i)	(ii)	(iii)	(iv)
(a)	X, Y	V	W, Z	Z
(b)	X, Y, Z	V, W	V, W, X, Z	Y
(c)	V, W, X, Z	V, W	Y	X, Y, Z
(d)	X, Y, Z	V, W	V, W	Y, Z

An

57. Observe the given figure carefully and choose the incorrect statement(s).



I. Large volume of acid should be added to a small volume of water slowly.
II. The figure represents the process of concentrating dilute H_2SO_4.
III. The process involves release of a large amount of heat.
IV. Water should not be stirred.

(a) I and IV only (b) II and III only
(c) I, II and IV only (d) II only

Ap

58. Seema conducted an experiment to test the nature of four different samples. She noted down her observations in the given table.

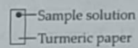
Sample	Effect on Turmeric solution	China rose indicator	Methyl orange indicator
W	Red	Green	Yellow
X	Yellow	Magenta	Red
Y	Yellow	Light pink	Orange
Z	Red	Green	Yellow

Which of the following conclusions can be drawn from this experiment?

(a) W and Z are basic, X is acidic and Y is neutral in nature.
(b) W is acidic, X is basic, Y and Z are neutral in nature.
(c) W and Z are acidic, X is basic and Y is neutral in nature.
(d) Y is neutral, W and X are basic and Z is acidic in nature.

An

59. Smita took a spoon of turmeric powder and made a paste by adding little water to it. She cut thin strips of a paper and applied this paste on them. After drying she put few drops of given sample solutions (i)-(iv) on these strips. What would be the changes in the colour she observed?

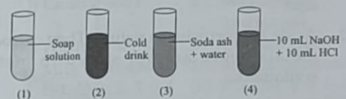


(i) *Amla* juice
(ii) Lime water
(iii) Common salt solution
(iv) Baking soda solution

(a) (i)-Yellow, (ii)-Red, (iii)-Yellow, (iv)-Red
(b) (i)-Red, (ii)-Yellow, (iii)-Yellow, (iv)-Red
(c) (i)-Red, (ii)-Red, (iii)-Yellow, (iv)-Yellow
(d) (i)-Yellow, (ii)-Yellow, (iii)-Red, (iv)-Red

An

60. During the science activity, class teacher arranged the test tubes as shown in the figure and asked students to predict the effect of various indicators on the given solutions.



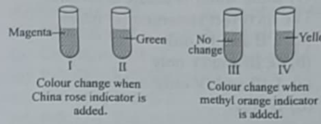
Which of the following statements are correct?

I. Turmeric solution will turn red in test tubes 1 and 4.
II. Methyl orange will turn yellow in test tubes 1 and 3.
III. China rose indicator will turn dark pink in test tube 2.
IV. Blue litmus solution will turn red in test tubes 1 and 4.

(a) I and IV only
(b) I, II and IV only
(c) II and III only
(d) None of these

An

61. Shruti tested the nature of a few common substances with the help of some indicators. Indicators used and changes she observed are as follows :



Substances present in test tubes I, II, III and IV are respectively

(a) salt solution, curd, lemon juice and vinegar
(b) orange juice, window cleaner, sugar solution and lime water
(c) vinegar, soda water, soap solution and lemon juice
(d) lemon juice, lime water, window cleaner and soap solution.

62. Fill in the blanks in the given table by choosing an appropriate option.

Solution	Indicators: Methyl orange	Phenolphthalein	Red rose
Common salt	No change	(i)	No change
Lime water	Yellow	Pink	(ii)
(iii)	Red	Colourless	(iv)
(v)	Yellow	Pink	Green

	(i)	(ii)	(iii)	(iv)	(v)
(a)	Red	Red	Spinach juice	Yellow	Sugar solution
(b)	Pink	Yellow	Salt solution	Pink	Lime water
(c)	Colourless	Green	Tomato juice	Red	Caustic soda
(d)	Yellow	Green	Salt solution	Yellow	Aerated drink

An

63. Consider the following statements :

I. Baking soda is an acidic salt whereas common salt is a basic salt.

II. The water extract of spinach does not change the colour of red litmus solution.
III. Organic matter neutralises the acidic nature of the soil.
IV. Lime water is acidic in nature.

The incorrect statement(s) is/are
(a) I, II and IV only
(b) II, III and IV only
(c) I, III and IV only
(d) II only.

64. Deepak found the soil of his field too basic in nature while Karan used excessive chemical fertilisers in his fields. What should they add to their fields to improve the quality of the soil?
(a) Deepak : Quick lime
Karan : Organic matter
(b) Deepak : Organic matter
Karan : Quick lime
(c) Both should add quick lime.
(d) Both should add organic matter. Ap

65. "We have to get the problem of acid rain under control. We must do whatever it takes to get the pH down to zero". The quote is
(a) absolutely correct
(b) wrong
(c) meaningless because pH of rain water has no relation with its acidic nature
(d) quite meaningful.

Fill in the Blanks

1. Bases which are soluble in water are called ______.
2. Substances used for testing acids and bases are called ______.
3. Bases react with acids to form ______ and ______.
4. Acids turn ______ litmus to ______ while bases turn ______ litmus to ______.
5. Phenolphthalein is ______ in acidic and ______ in basic solutions.
6. Acidity or indigestion in stomach is due to excessive secretion of ______.
7. ______ acids dissociate completely in water while ______ acids do not dissociate completely.
8. Acids which are found in plants and animals are called ______ acids and acids which are prepared artificially from minerals are called ______ acids.
9. Substances that are neither acidic nor basic are called ______.
10. With some samples under test, the China rose turns magenta or dark pink. Such samples are ______ in nature.
11. Turmeric paper indicator remains yellow in some solutions. Such solutions are ______ or ______ in nature.
12. ______ and ______ are examples of two natural indicators.
13. Sodium hydroxide is a ______ base.
14. Milk of magnesia contains ______.
15. ______ can be prepared by neutralisation reaction between an acid and a base.

True or False

1. Chloride salts are obtained from hydrochloric acid.
2. Acids are corrosive in nature. They corrode metals hence are not stored in metal containers.
3. Litmus is an indicator prepared by dissolving litmus salt in water.
4. Antacids neutralise the bases present in the stomach.
5. When an acid reacts with a base, neutralisation reaction takes place to give salt and water.
6. Acids have a sour taste and they are soapy to touch.
7. All bases are alkalies but all alkalies are not bases.
8. Indicators are substances that are used to identify whether a compound is acidic, basic or neutral.

9. Neutral substances do not bring about any change in colour of indicators.
10. The colour of phenolphthalein is pink in acidic solutions while colourless in basic solutions.
11. Sodium bicarbonate is used in preparing bakery products.
12. The acids which dissociate completely in water to furnish large number of hydrogen ions are called strong acids.
13. Acidity is caused by excessive secretion of acetic acid in stomach.
14. Ammonium hydroxide is a base present in window cleaners.
15. Common salt is used in preservation of pickles.

Match the Following

In this section, each question has two matching lists. Choices for the correct combination of elements from List-I and List-II are given as options (a), (b), (c) and (d) out of which one is correct.

1.

List-I	List-II
(P) Quick lime	1. Sodium hydroxide
(Q) Caustic soda	2. Magnesium hydroxide
(R) Washing soda	3. Calcium oxide
(S) Milk of magnesia	4. Sodium carbonate

(a) P-1, Q-2, R-3, S-4 (b) P-3, Q-1, R-4, S-2
(c) P-2, Q-3, R-4, S-1 (d) P-4, Q-2, R-1, S-3

2.

List-I	List-II
(P) Citric acid	1. Tamarind
(Q) Lactic acid	2. Spinach
(R) Oxalic acid	3. Orange
(S) Tartaric acid	4. Sour milk

(a) P-3, Q-4, R-2, S-1 (b) P-4, Q-3, R-2, S-1
(c) P-1, Q-3, R-4, S-2 (d) P-2, Q-1, R-3, S-4

3.

List-I	List-II
(P) Sulphuric acid	1. Weak acid
(Q) Sodium hydroxide	2. Strong acid
(R) Ammonium hydroxide	3. Strong base
(S) Acetic acid	4. Weak base

(a) P-3, Q-2, R-4, S-1 (b) P-4, Q-3, R-1, S-2
(c) P-2, Q-3, R-4, S-1 (d) P-1, Q-3, R-2, S-4

4.

List-I	List-II
(P) Water soluble base	1. Sodium chloride
(Q) Organic acid	2. Sulphuric acid
(R) Mineral acid	3. Alkali
(S) Common salt	4. Acetic acid

(a) P-3, Q-4, R-2, S-1
(b) P-4, Q-3, R-2, S-1
(c) P-2, Q-1, R-4, S-3
(d) P-1, Q-2, R-4, S-3

5.

List-I	List-II
(P) Ant sting	1. Vinegar
(Q) Wasp sting	2. Milk of magnesia
(R) Antacid	3. Calamine solution
(S) Acidic soil	4. Slaked lime

(a) P-4, Q-2, R-3, S-1 (b) P-1, Q-3, R-4, S-2
(c) P-3, Q-1, R-2, S-4 (d) P-2, Q-4, R-1, S-3

Assertion & Reason Type

Directions : In the following questions, a statement of assertion is followed by a statement of reason. Mark the correct choice as :

(a) If both assertion and reason are true and reason is the correct explanation of assertion.
(b) If both assertion and reason are true but reason is not the correct explanation of assertion.
(c) If assertion is true but reason is false.
(d) If both assertion and reason are false.

1. **Assertion :** The pain caused due to ant sting can be neutralised by baking soda.
Reason : Ant sting is acidic in nature.

2. **Assertion :** A salt is produced when an acid is neutralised by a base.
Reason : A salt can be acidic, basic or neutral.

3. **Assertion :** To neutralise the excess acid formed in the stomach, milk of magnesia is taken.
Reason : Milk of magnesia contains a base called magnesium hydroxide.

4. **Assertion** : With some samples turmeric paper turns red.
Reason : Such samples are acidic in nature.

5. **Assertion** : Methyl orange and phenolphthalein are natural indicators.
Reason : Methyl orange is yellow in colour while phenolphthalein is pink.

6. **Assertion** : Lime water turns red rose indicator into green.
Reason : Lime water is basic in nature.

7. **Assertion** : pH of acetic acid is less than 7.
Reason : Acetic acid is an organic acid.

8. **Assertion** : NaCl is neutral salt.
Reason : It is combination of strong acid and strong base.

9. **Assertion** : Salt and water are formed in neutralisation reaction.
Reason : Heat is released in a neutralisation reaction.

10. **Assertion** : Odour of olfactory indicator change in an acidic or basic medium.
Reason : Clove oil is an olfactory indicator.

Comprehension Type

PASSAGE-I : The process in which an acid completely reacts with a base to form salt and water is called neutralisation. The salts which are formed may be acidic, basic or neutral. The substances which do not bring any change in the colour of the indicators are called neutral substances.

1. When sodium hydroxide is added to dilute hydrochloric acid
(a) only sodium chloride salt is formed.
(b) sodium chloride salt and water are formed and no heat is evolved.
(c) sodium chloride salt and water are formed and heat is evolved.
(d) sodium chloride salt and water are formed and heat is absorbed.

2. When a few drops of phenolphthalein are added to a solution of common salt
(a) the solution becomes pink in colour
(b) the solution remains colourless
(c) the solution turns orange in colour
(d) the solution turns green in colour.

3. Sodium chloride solution is neutral in nature because
(a) it is formed by complete neutralisation of a strong acid and a strong base.
(b) it is formed by the complete neutralisation of a weak acid and a weak base.
(c) it is formed by the incomplete neutralisation of a strong acid and a strong base.
(d) it is formed by the incomplete neutralisation of a weak acid and a weak base.

PASSAGE-II : Acids which dissociate completely in water to give a large number of hydrogen ions are called strong acids while the acids which do not dissociate completely in water are called weak acids. Acids can also be classified as mineral acids which are prepared from minerals and organic acids which are found in plants and animals.

1. Sulphuric acid and hydrochloric acid are examples of
(a) strong mineral acids
(b) weak mineral acids
(c) strong organic acids
(d) weak organic acids.

2. Which of the following groups of acids contains weak organic acids?
(a) Acetic acid, Formic acid, Nitric acid
(b) Acetic acid, Carbonic acid, Formic acid
(c) Hydrochloric acid, Sulphuric acid, Formic acid
(d) Citric acid, Lactic acid, Sulphuric acid

3. Which of the following will turn blue litmus red – (i) hydrochloric acid, (ii) vinegar, (iii) lemon juice, (iv) orange juice?
(a) Only (i)
(b) (ii), (iii) and (iv)
(c) (i), (ii) and (iii)
(d) (i), (ii), (iii) and (iv)

PASSAGE-III : Acids are sour in taste and bases are bitter in taste. Acids and bases both are quite useful in our daily life. Many edible items contain acids which are not very strong ones but weak acids.

1. The acids present in curd and spinach are respectively
(a) lactic acid and citric acid
(b) citric acid and oxalic acid
(c) oxalic acid and acetic acid
(d) lactic acid and oxalic acid.

2. The acid present in our body which is responsible to pass characters from one generation to another is
(a) ascorbic acid
(b) deoxyribonucleic acid
(c) tartaric acid
(d) lactic acid.

3. How many of the given substances are basic in nature?
Vinegar, Common salt, Toothpaste, Soap, Grapes
(a) 2 (b) 3
(c) 4 (d) 5

4. Bases present in milk of magnesia and lime water respectively are
(a) sodium hydroxide, magnesium hydroxide
(b) magnesium hydroxide, sodium hydroxide
(c) ammonium hydroxide, potassium hydroxide
(d) magnesium hydroxide, calcium hydroxide.

PASSAGE-IV : Neutralisation reactions are used to treat many daily life problems like acidity, bee and wasp stings, tooth decay, etc. These are also used in treatment of factory waste, sewage waste and also quite useful in agriculture.

1. Which of the following statements is/are not correct?
(a) Organic matter can be used to treat basic soil.
(b) Vinegar can be used to treat wasp sting.
(c) Slaked lime can be used to treat indigestion.
(d) All of these.

2. Fill in the blanks by choosing an appropriate option.
When an ant bites, it injects ______ into the skin, the effect of this can be neutralised by rubbing moist ______ or ______.
(a) base, vinegar, curd
(b) base, milk of magnesia, vinegar
(c) acid, milk of magnesia, vinegar
(d) acid, baking soda, calamine solution

3. When the soil is too acidic, it is treated with
(a) quick lime (b) slaked lime
(c) organic matter (d) either (a) or (b).

4. Which of the following represents to a neutralisation reaction?
(a) $HCl + NaOH \longrightarrow NaCl + H_2O$
(b) $HCl + H_2O \longrightarrow H_3O^+ + Cl^-$
(c) $SO_2 + H_2O \longrightarrow H_2SO_3$
(d) $NaOH + H_2O \longrightarrow Na^+ + OH^-$

Subjective Problems

Very Short Answer Type

1. Name the acid present in vinegar.
2. Give names of any two natural indicators.
3. What is a neutral salt?
4. Name the acid and the base from which the salt calcium chloride is formed.
5. Can we taste acids and bases to identify them?
6. What is the colour of phenolphthalein in basic solution?
7. Which chemical is used to neutralise the acidic soil?
8. Name the base which is present in a window cleaner.
9. Write the chemical name and formula of baking soda.
10. Name one strong and one weak acid.

11. Name two synthetic acid-base indicators.
12. Name the acid present in grapes.
13. What are those substances called which change colour with acids or bases?
14. What is the reaction between an acid and a base called?
15. Which acid is present in sour milk?

Short Answer Type

1. (a) What is an indicator?
   (b) How will you prepare China rose indicator?
2. Give an important use of neutralisation reaction in daily life.
3. (a) Why do doctors advice to make up for salt loss to the people who sweat more?
   (b) Give the colours of phenolphthalein and methyl orange indicators in acidic and basic media.
4. A substance '*X*' gives pink colour with phenolphthalein. When another substance '*Y*' is added to it, the pink colour disappears after some time. What are substances '*X*' and '*Y*'?
5. (a) Which acids are used to make the following salts?
   (i) Sulphates
   (ii) Acetates
   (iii) Nitrates
   (b) Why should curd and other sour substances not be kept in aluminium or copper containers?
6. What are strong and weak bases? Give two examples of each.
7. Write important uses of hydrochloric and sulphuric acids.
8. Are all acids corrosive in nature? Name few acids which are non-corrosive and may be part of our food.
9. (a) Why do antacid tablets relieve the uneasiness due to indigestion?
   (b) What is used to neutralise an excessive basic soil?
10. (a) Why is sulphuric acid called "King of Chemicals"?
    (b) Which bases are called alkalies? Give an example of alkali.

Long Answer Type

1. Four solutions *A*, *B*, *C* and *D* are taken in four test tubes. They were tested with four indicators—methyl orange, phenolphthalein, China rose and blue litmus. The observations are recorded below. Identify *A*, *B*, *C* and *D* as acids, bases or neutral salts.

Soln.	Methyl orange	Phenol-phthalein	China rose	Blue litmus
A	Orange	Colourless	Pink	Blue
B	Yellow	Pink	Green	Blue
C	Red	Colourless	Magenta	Red
D	Yellow	Pink	Green	Blue

2. What is meant by acid rain? How is acid rain caused? What damage is caused by acid rain?
3. Give few important applications of neutralisation reactions.
4. (a) How will you prepare turmeric indicator? Using this indicator give tests for lime water, salt solution and soap solution.
   (b) Can turmeric paper be used to differentiate between an acid and a neutral solution? Explain your answer.
5. (a) Complete the following table :

Indicator	Colour in acidic medium	Colour in basic medium	Colour in neutral medium
Blue litmus	____	____	____
Red litmus	____	____	____
China rose	____	____	____
Turmeric	____	____	____
Methyl orange	____	____	____
Phenolphthalein	____	____	____

   (b) Why do we feel a burning sensation in the stomach when we overeat? What is the medicine used? Given one example.

6. (a) Give one word answer for the following:
   (i) A base which is soluble in water.
   (ii) A compound which changes colour with acid and base.
   (iii) An indicator extracted from lichens.
   (iv) A substance formed when an acid and a base react.
   (v) A substance which is sour in taste and gives hydrogen ions in the solution.
   (b) If we touch the test tube immediately after carrying out the neutralization reaction of an acid and base in it, it is found to be hot. Explain.
7. You are given five samples *A*, *B*, *C*, *D* and *E*. Out of these, two are acidic, two are basic and one is neutral. (*D*) is a stronger acid than (*C*). (*B*) is neutral while (*E*) is a weaker base than (*A*). Match these samples with the given pH values: 7.0, 12.8, 2.1, 9.4 and 5.5? What is the significance of pH scale?
8. Tooth enamel is one of the hardest substance in our body. How does it damage due to eating chocolates and sweets? What should we do to prevent it?
9. What will be the action of the following substances on litmus paper?
   Dry HCl gas, moistened NH_3 gas, lemon juice, carbonated soft drink, curd, soap solution.
10. (a) A yellow curry stain on a white shirt turns red when it is washed with soap. Give reason.
    (b) Dorji has a few bottles of soft drink in his restaurant. But unfortunately, these are not labelled. He has to serve the drinks on the demand of customers. One customer wants acidic drink, another wants basic and third one wants neutral drink. How will Dorji decide which drink is to be served to whom?

Integer/Numerical Value Type

1. You are provided with the following substances in laboratory :
   Hydrochloric acid, sulphuric acid, sodium hydroxide, potassium hydroxide, ammonium hydroxide.
   How many substances will give green colour with red rose indicator?
2. How many of the following samples are neutral?
   Sugar solution, shampoo, window cleaner, salt solution, distilled water, potassium nitrate solution.
3. How many samples will give red colour with turmeric?
   Curd, tamarind, milk, baking soda, lime water.
4. pH of a neutral solution is _______.
5. How many types of salts can be formed as a result of neutralisation?

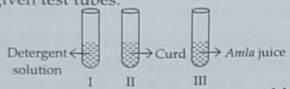
Case Based Questions

Case I : Aayan took few petals of China rose in a beaker. He added some water and heated the beaker till the water in it became warm. He kept the mixture for some time till the water became coloured. He tested the given samples with China rose indicator to find the nature of the samples. The given samples were lemon juice, soap solution, distilled water, sugar solution, lime water, curd, window cleaner.

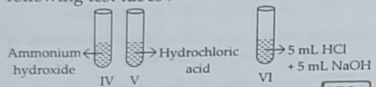
1. With some samples the China rose indicator turned dark pink or magenta. Such samples are
   (a) basic
   (b) acidic
   (c) neutral
   (d) both acidic and neutral.
2. The indicator turned green colour with
   (a) soap solution, lime water and window cleaner
   (b) lemon juice, distilled water, sugar solution
   (c) lime water, curd, window cleaner
   (d) soap solution, lemon juice, sugar solution.
3. Which of the samples did not show any change in colour with China rose indicator?
   (a) Lemon juice, curd and sugar solution
   (b) Distilled water and sugar solution

(c) Distilled water, window cleaner and lime water
(d) Lime water, soap solution and lemon juice

Case II : Red rose indicator has been added to the given test tubes.



Phenolphthalein indicator has been added to following test tubes :



CFQs

1. Colour change in the given test tubes will be

	I	II	III
(a)	Light pink	Red	Light green
(b)	Green	Red	Red
(c)	Magenta	Green	Green
(d)	Green	Light pink	Red

2. The colour of solution in test tubes IV, V and VI will be
(a) pink, pink, colourless
(b) pink, colourless, colourless
(c) pink, pink, green
(d) colourless, pink, colourless.

3. Complete the given table by choosing correct option.

Test tube	Effect on blue litmus paper	Effect on turmeric paper
I	No change	Y
V	X	No change

	X	Y
(a)	Red	Yellow
(b)	Red	Blue
(c)	Red	Red
(d)	No change	Red

4. How many of the given substances (I–VI) will turn red litmus to blue?
(a) 2 (b) 3
(c) 4 (d) 6

Acids → Hydrogen$^+$ ions
Bases → Hydroxide$^-$ ions.

11/8/26.

The World of Metals and Non-metals

CHAPTER 02

Learning Outcomes...

- Metals
- Physical and Chemical Properties of Metals and Uses
- Corrosion of Metals
- Prevention of Corrosion
- Non-metals
- Physical and Chemical Properties of Non-metals and Uses
- Comparison of Physical and Chemical Properties of Metals and Non-metals

Introduction

Elements have been broadly divided into two main groups depending on their properties, metals and non-metals. Besides these, there are some elements which show the properties of both metals and non-metals, they are called **metalloids**. All elements are arranged in the periodic table according to their properties. Presently 118 elements are known. These elements are the basic building blocks of all matter. Some are naturally occurring while others are artificially made in the laboratory, and do not exist in nature.

Metals

Metals are those elements which give electrons to form positive ions or cations, *i.e.*, they are electropositive in nature. Generally they contain 1, 2, or 3 valence electrons.
Few metals occur in free or native state, *e.g.*, gold, silver and platinum. They are called **noble metals**. Reactive metals like sodium, aluminium, zinc, etc., occur in combined state in the form of chemical substances called **minerals**. The minerals from which metals can be extracted conveniently and profitably are called **ores**. The process of obtaining a pure metal from one of its ore is known as **metallurgy.**

Physical Properties of Metals

Metals can be distinguished from non-metals on the basis of their physical and chemical properties.

- **Physical state :** Almost all metals are solids at room temperature (except mercury and gallium which are liquids) at room temperature.
- **Melting and boiling points :** Melting and boiling points of metals are very high (except mercury and gallium).
- **Hardness :** Metals are generally hard (except lithium, sodium and potassium which are soft metals and can be cut with a knife).
- **Lustre :** Metals have a shine of their own and can also be polished. Lustre shown by metals is known as metallic lustre.
- **Malleability and ductility :** The property of metals by which they can be beaten into thin sheets is called **malleability** and the property of metals by which they can be drawn into thin wires is called **ductility**. Metals possess these properties in varying proportions because their atoms are arranged in layers, so they can slip past each other when hammered or pulled. Gold and silver are the most malleable metals.

DO YOU KNOW? Gold is so ductile that one gram of it can be drawn into a 2-kilometre-long wire approximately.